

NGUYEN MINH DUC

☎ +84 363 715 481 | ✉ ducnguyenm810@gmail.com | 🔗 LinkedIn | 🌐 GitHub | 📍 Thanh Xuan, Ha Noi

EDUCATION

VinUniversity (VinUni)

M.Sc. Computer Science – Full Scholarship

Hanoi, Vietnam

2026–Present

Advisor: **Asst. Prof. Le Duy Dung**

Hanoi University of Science and Technology (HUST)

B.Sc. Mathematics and Informatics

Hanoi, Vietnam

2019–2024

Advisor: **Dr. Tran Ngoc Thang**

RESEARCH INTERESTS

Multi-Objective Optimization and Learning; Pareto Front Learning; Hypernetwork-Based Learning; Multi-Objective Recommender Systems; Large Language Models and Agentic Recommendation.

PUBLICATIONS

Nguyen, M.D. & Le, D.D. (2026). *A Framework for Controllable Multi-objective Learning with Annealed Stein Variational Hypernetworks*. ACM Transactions on Intelligent Systems and Technology (TIST).

Nguyen, M.D., Dinh, P.M., Nguyen, Q.H., Long, H.P., & Le, D.D. (2025). *Improving Pareto Set Learning for Expensive Multi-objective Optimization via Stein Variational Hypernetworks*. Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2025), 39(18), 19677–19685.

Kieu, H.D., **Nguyen, M.D.**, Buntine, W., Nguyen, T.-S., & Le, D.D. (2026). *KALM4Rec: Keyword-Driven Retrieval-Augmented Large Language Models for Cold-Start User Recommendations*. *Machine Learning*, 115, 142.

Kieu, H.D., Zhang, D.C., **Nguyen, M.D.**, Wu, Q., Xu, M., & Le, D.D. (2026). *PromptHG: Prompt-Enhanced Heterogeneous Graph for Personalized News Recommendation*. The 48th European Conference on Information Retrieval (ECIR 2026).

Kieu, H.D., Zhang, D.C., **Nguyen, M.D.**, Xu, M., Wu, Q., & Le, D.D. (2025). *Enhancing News Recommendation with Hierarchical LLM Prompting*. In Companion Proceedings of the ACM Web Conference (WWW 2025).

Kieu, H.D., **Nguyen, M.D.**, Nguyen, T.S., & Le, D.D. (2025). *Keyword-driven Retrieval-Augmented Large Language Models for Cold-start User Recommendations*. In Companion Proceedings of the ACM Web Conference (WWW 2025).

Nguyen, M.D., Le, D.D., & Nguyen, P.L. (2025). *Optimizing Electric Vehicle Charging Station Placement Using Reinforcement Learning and Agent-Based Simulations*. Preprint.

Nguyen, M.D., Kieu, H.D., & Le, D.D. (2026). *AMEM4Rec: Leveraging Cross-User Similarity for Memory Evolution in Agentic LLM Recommenders*. Preprint.

EXPERIENCE

R2Studio, VinUniversity

Research Assistant; Advisor: **Asst. Prof. Le Duy Dung**

Sep 2023 – Present

- Conducted research on multi-objective optimization, Pareto front learning, hypernetworks, and recommender systems.
- Developed optimization methods based on Stein Variational Gradient Descent (SVGD) and Hypernetworks for Pareto set learning.
- Investigated large language model (LLM)-based and agentic recommender systems for personalized and cold-start recommendation.

AI for Materials Science: Inverse Design for Beam

Advisor: **Asst. Prof. Le Duy Dung**; **Asst. Prof. Nguyen Dang Tung**

Sep 2024 – 2026

- Developed a generative model for synthetic data augmentation and novel material design exploration.

- Applied multi-objective Bayesian optimization to identify beam designs that balance multiple structural objectives.
- Evaluated the proposed framework for efficient exploration of the material design space.

Electric Vehicle Charging Station Placement via Reinforcement Learning

Advisor: **Asst. Prof. Le Duy Dung; Dr. Nguyen Phi Long**

Sep 2024 – May 2025

- Developed a reinforcement learning (RL) framework for optimizing charging-station locations with a tailored action space.
- Integrated an agent-based simulation environment to model urban EV mobility and provide feedback for sequential charging-station placement decisions.
- Demonstrated improved placement efficiency compared with traditional heuristic approaches.

Multi-objective Bayesian Optimization

Advisor: **Asst. Prof. Le Duy Dung**

Sep 2023 – 2024

- Developed a Stein Variational Gradient Descent (SVGD) method for expensive multi-objective optimization.
- Combined SVGD with Hypernetworks to improve Pareto set coverage and balance exploration–exploitation trade-offs.
- Validated the proposed method through extensive experiments against conventional multi-objective optimization baselines.

REFERENCES

Asst. Prof. Le Duy Dung

College of Engineering and Computer Science, VinUniversity

Website

Email: dung.ld@vinuni.edu.vn

Dr. Tran Ngoc Thang

Faculty of Applied Mathematics and Informatics, Hanoi University of Science and Technology

Google Scholar

Email: thang.tranngoc@hust.edu.vn